



24-

mavzu

MASALALAR YECHISH

1 Suv bilan to'ldirilgan idish tubida yuzasi 20 cm² bo'lgan teshik bor. Teshikning markazi suv sathidan 2 m pastda. Teshik probka bilan mahkamlab qo'yilgan. Probkaga ta'sir etuvchi gidrostatik bosim kuchi nimaga teng? Suvning zichligi 1000 kg/m³.

Berilgan:	Formula	Hisoblash
$S = 20 \text{ cm}^2 = 0,002 \text{ m}^2$ $h = 2 \text{ m}$ $\rho = 1000 \frac{\text{kg}}{\text{m}^3}$ $g = 9,81 \frac{\text{N}}{\text{kg}}$	$p = \rho g h$ $[p] = \frac{\text{kg}}{\text{m}^3} \cdot \frac{\text{N}}{\text{kg}} \cdot \text{m} = \frac{\text{N}}{\text{m}^2} = \text{Pa}$ $p = \frac{F}{S} \quad F = p \cdot S$	$p = 1000 \cdot 9,81 \cdot 2 \text{ Pa} = 19620 \text{ Pa}$ $F = 19620 \cdot 0,002 \text{ N} = 39,24 \text{ N}$ Javob: $F = 39,24 \text{ N}$.
Topish kerak: $F = ?$	$[F] = \text{Pa} \cdot \text{m}^2 = \frac{\text{N}}{\text{m}^2} \cdot \text{m}^2 = \text{N}$	

2 Tutash idishlarda qanday balandlikdagi kerosin ustuni 2,5 cm balandlikdagi simob ustunini muvozanatlay oladi? Bunda kerosin zichligini 800 kg/m³, simob zichligini 13600 kg/m³ ga teng deb oling.

Berilgan:	Formula	Hisoblash
$\rho_k = 800 \text{ kg/m}^3$ $\rho_s = 13600 \text{ kg/m}^3$ $g = 9,81 \frac{\text{N}}{\text{kg}}$ $h_s = 2,5 \text{ cm} = 0,025 \text{ m}$	$p_1 = p_2$ $\rho_k g h_k = \rho_s g h_s$ $\frac{h_k}{h_s} = \frac{\rho_s}{\rho_k}$ $h_k = \frac{\rho_s}{\rho_k} \cdot h_s$	$h_k = \frac{13600}{800} \cdot 0,025 \text{ m} = 0,425 \text{ m}$ Javob: $h_k = 42,5 \text{ cm}$.
Topish kerak: $h_k = ?$	$[h] = \frac{\frac{\text{kg}}{\text{m}^3}}{\frac{\text{kg}}{\text{m}^3}} \text{ m} = \text{m}$	

3 Idish tubida yuzasi 24 cm^2 bo'lgan teshik tiqin bilan berkitilgan. Bu tiqin 12 N kuch ta'sirida teshikdan tashqariga sug'uriladi. Tiqin chiqib ketmasligi uchun idishga qanday balandlikda suv quyish kerak?

Berilgan:	Formula	Hisoblash
$F = 12 \text{ N}$ $S = 24 \text{ cm}^2 = 0,0024 \text{ m}^2$ $\rho_s = 1000 \text{ kg/m}^3$ $g = 9,81 \frac{\text{N}}{\text{kg}}$	$p = \frac{F}{S}$ $p = \rho g h$ $\rho g h = \frac{F}{S}$ $h = \frac{F}{\rho g S}$ $[h] = \frac{\text{N}}{\frac{\text{kg}}{\text{m}^3} \frac{\text{N}}{\text{kg}} \cdot \text{m}^2} = \text{m}$	$h = \frac{12}{1000 \cdot 9,81 \cdot 0,0024} \text{ m} \approx 0,51 \text{ m}$ Javob: $h \approx 51 \text{ cm}$.
Topish kerak: $h = ?$		



16-mashq

1 Suv minorasining tubiga o'rnatilgan monometr 147 kPa bosimni ko'rsatsa, minoradagi suv sathining balandligini aniqlang.

2 Idishlarning biriga spirt, ikkinchisiga esa simob quyilgan. Idishlardagi spirt va simobning balandligi 8 cm ga teng. Ularning idish tubiga beradigan gidrostatik bosimlari farqini hisoblang.

3 Suyuqlik ustunining o'zgarishiga ko'ra uning idish tubiga beradigan bosimi grafik ko'rinishda tasvirlangan. Grafikdan foydalanib suyuqlik turini aniqlang. (g ning qiymatini 10 N/kg deb oling).

4 Akvarium tubiga va baliqqa ta'sir qilayotgan gidrostatik bosimni aniqlang.

